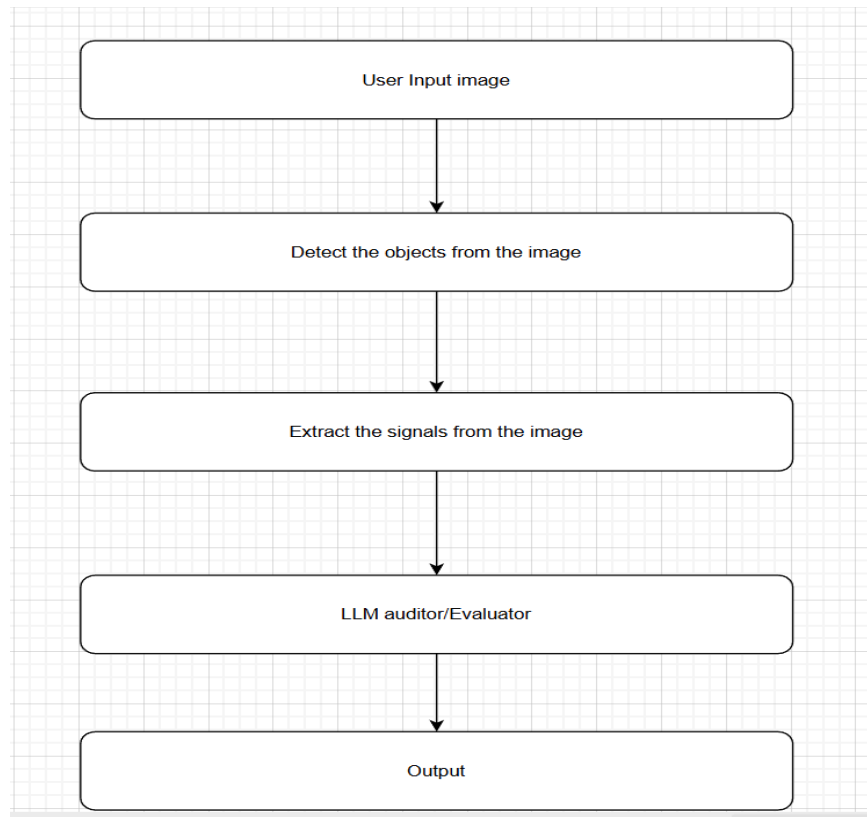


System architecture



Instead of a "black-box" end-to-end model, the system breaks the audit into discrete, verifiable steps:

- **Stage 1: Object Localization (YOLOv12n):** Used to identify the primary product and generate bounding boxes. This provides the spatial context needed for localized analysis.
- **Stage 2: Signal Extraction (OpenCV):**
 - * **Sharpness:** Calculates Laplacian variance on the product crop vs. the global image.
 - **Composition:** Analyzes centering, object coverage (surface area), and lighting (pixel intensity).
- **Stage 3: LLM Auditor (DeepSeek-R1-8B):** A reasoning-focused LLM that synthesizes these raw signals into a human-readable verdict.
- **Stage 4: Structured Output:** Utilizing Pydantic to ensure the final output is a machine-readable JSON object, ready for integration into e-commerce databases.

Why you chose your approach

- **Deterministic Heuristics over Multimodal Guessing:** By calculating sharpness and lighting in Python/OpenCV, we provide the LLM with "ground truth" facts. This prevents the LLM from hallucinating focus quality based on the "vibe" of the image.

- **Localized vs. Global Analysis:** A global sharpness score can be misleading (e.g., a sharp background with a blurry product). Our system isolates the product crop, allowing for a "Subject-to-Background" comparison that identifies professional bokeh.
- **Local LLM (Ollama):** Choosing DeepSeek-R1-8B via Ollama ensures data privacy (images don't leave the local server) and zero API latency/cost for the reasoning step.

Limitations and Improvements

There are some areas which can be improved

- **Texture Bias:** The Laplacian method depends on edges. Minimalist products (e.g., a smooth white bowl) may return low sharpness scores despite being in focus. **Improvement:** Implement a deep-learning-based "No-Reference" Quality Assessment (NR-IQA) model.
- **Contextual Aesthetic Intent:** A "Rule of Thirds" professional shot might be flagged as "Not Centered" by our current geometric check. **Improvement:** Integrate an aesthetic scoring model to distinguish between "Off-center Error" and "Artistic Composition."
- **Lighting Complexity:** Global pixel mean doesn't account for harsh shadows or high dynamic range. **Improvement:** Use histogram analysis to detect "clipping" in highlights or shadows.

How to Productionize This

- **Model Optimization:** Convert the YOLO model to **TensorRT** or **OpenVINO** for sub-millisecond inference on edge devices or servers.
- **Scalability:** Move the LLM reasoning to an asynchronous task queue (like **Celery/RabbitMQ**) to handle high-volume uploads without blocking the UI.
- **CI/CD & Monitoring:** Implement a "Ground Truth" with 100 labeled "Sharp" and "Blurry" images. Any change to the code or thresholds must pass this test to ensure no regressions in audit quality.
- **Model Drift Tracking:** Monitor how often the LLM disagrees with the OpenCV signals. If the LLM frequently rejects images with high sharpness scores, it's a signal that the sharpness threshold needs recalibration.